Title: A novel orientation strategy for human post-mortem organotypic eye culture

Authors:
Fleur Schipper[1,3], Bonnie C. Plug[1,2,3], Zoë A.R. Keasberry[2], Wilma D.J. van de Berg[2, 3], Jurre den Haan[4], Yolande A. L. Pijnenburg[4], Marianna Bugiani[1,3].

1 Department of Pathology, Amsterdam University Medical Center, Amsterdam, The Netherlands
2 Department of Anatomy and Neurosciences, Section Clinical Neuroanatomy and Biobanking, Amsterdam University Medical Center, The Netherlands
3 Amsterdam Neuroscience, program Neurodegeneration, Amsterdam, The Netherlands
4 Department of Neurology, Amsterdam University Medical Center, The Netherlands

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Background

Current human organotypic retinal cultures (HORC) models utilize detached retinas and are limited to flatmount or trephine punched tissue orientation which impacts physiological polarity and influences cellular responses as glial activation and protein distribution.

Objective

We propose a novel organotypic eye culturing model that contains the retina, retinal pigment epithelium-choroid, sclera and optic nerve to preserve native anatomical context, maintain retinal integrity and improve structural preservation and experimental fidelity.

Methods:

Post-mortem human eyes will be collected within 6 hours post-mortem and dissected using a modified orientation approach. The anterior part of the eye between lens and ora serrata will be removed, leaving a residual tissue block in the craniocaudal plane adjacent to the optic nerve on both nasal and temporal sides. 300µm-thick slices including the optic nerve will be cut with a vibratome and cultured onto semi-permeable membrane inserts up to three weeks. This set up is designed to include both inferior and superior retina, maintain eye globe tissue architecture, and ensure equal exposure to medium to all retinal cell layers. One organotypic slice will be immediately fixed as a reference, and the others later at 5 ,11 and 18 days in vitro. Tissue integrity and viability will be assessed using histology, immunohistochemistry, and a life-death assay.

Expected results:
We previously developed a similar approach to culture ex vivo post-mortem human brain organotypic slice cultures. We hypothesize that this novel approach can be successfully adapted to human organotypic eye cultures resulting in improved preservation of retinal architecture and more physiologically relevant multi-cellular responses.

Conclusion:
This novel orientation strategy may enhance the utility of HORCs by better preserving native anatomical context. As such, it could provide a more physiologically relevant experimental platform to study disease development and test potential therapies.